Examples of Weak Derivatives

Recall that a function $w \in L^1_{\mathrm{loc}}(\Omega)$ is the *weak derivative* of $u \in L^1_{\mathrm{loc}}(\Omega)$ if

$$\int_\Omega w\varphi \,\mathrm{d}x = -\int_\Omega u \frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x \qquad \forall \varphi \in C_0^\infty(\Omega). \tag{1}$$

We then write

$$w = \frac{\mathrm{d}u}{\mathrm{d}x},$$

where the derivative is understood in the weak sense.

Example 1: The absolute-value function

Let

$$\Omega = (-1,1), \qquad u(x) = |x|.$$

The classical derivative of u does not exist at $x = 0$. Define

$$w(x) = \begin{cases} -1, & -1 < x < 0, \\ 1, & 0 < x < 1. \end{cases}$$

We claim that w is the weak derivative of u.

To prove the claim, let $\varphi \in C_0^\infty(-1,1)$. We must show that

$$\int_{-1}^1 w\varphi \,\mathrm{d}x = -\int_{-1}^1 u \frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x.$$

For the left-hand side,

$$\begin{aligned} \int_{-1}^1 w\varphi \,\mathrm{d}x &= \int_{-1}^0 (-1)\varphi \,\mathrm{d}x + \int_0^1 \varphi \,\mathrm{d}x \\ &= -\int_{-1}^0 \varphi \,\mathrm{d}x + \int_0^1 \varphi \,\mathrm{d}x. \end{aligned} \tag{2}$$

For the right-hand side, since

$$|x| = \begin{cases} -x, & -1 < x < 0, \\ x, & 0 < x < 1, \end{cases}$$

we have

$$\begin{aligned} -\int_{-1}^1 |x| \frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x &= -\int_{-1}^0 (-x)\frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x - \int_0^1 x\frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x \\ &= \int_{-1}^0 x\frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x - \int_0^1 x\frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x. \end{aligned} \tag{3}$$

Using the product rule,

$$x\frac{\mathrm{d}\varphi}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}(x\varphi) - \varphi,$$

so

$$
\begin{aligned}
-\int_{-1}^{1} |x| \frac{\mathrm{d}\varphi}{\mathrm{d}x}\, \mathrm{d}x &= \int_{-1}^{0} \left[\frac{\mathrm{d}}{\mathrm{d}x}(x\varphi) - \varphi \right] \mathrm{d}x - \int_{0}^{1} \left[\frac{\mathrm{d}}{\mathrm{d}x}(x\varphi) - \varphi \right] \mathrm{d}x \\
&= [x\varphi(x)]_{-1}^{0} - \int_{-1}^{0} \varphi\, \mathrm{d}x - [x\varphi(x)]_{0}^{1} + \int_{0}^{1} \varphi\, \mathrm{d}x. \qquad (4)
\end{aligned}
$$

Since $\varphi \in C_0^\infty(-1,1)$,

$$\varphi(-1) = \varphi(1) = 0.$$

The terms involving $\varphi(0)$ also vanish because they are multiplied by $x = 0$. Therefore,

$$[x\varphi(x)]_{-1}^{0} = [x\varphi(x)]_{0}^{1} = 0.$$

Hence,

$$-\int_{-1}^{1} |x| \frac{\mathrm{d}\varphi}{\mathrm{d}x}\, \mathrm{d}x = -\int_{-1}^{0} \varphi\, \mathrm{d}x + \int_{0}^{1} \varphi\, \mathrm{d}x.$$

Comparing with (2),

$$\int_{-1}^{1} w\varphi\, \mathrm{d}x = -\int_{-1}^{1} u \frac{\mathrm{d}\varphi}{\mathrm{d}x}\, \mathrm{d}x.$$

Thus, w is the weak derivative of u, and

$$\boxed{\frac{\mathrm{d}u}{\mathrm{d}x} = \begin{cases} -1, & -1 < x < 0, \\ 1, & 0 < x < 1, \end{cases} \qquad \text{a.e. in } (-1,1).}$$

The value assigned to the weak derivative at the single point $x = 0$ is irrelevant.

Example 2: Strain jump at a material interface

Weak derivatives arise naturally in solid mechanics. Consider a one-dimensional bar occupying

$$\Omega = (0, L)$$

and composed of two linear elastic materials joined at $x = a$, where $0 < a < L$. The Young's modulus is

$$E(x) = \begin{cases} E_1, & 0 < x < a, \\ E_2, & a < x < L, \end{cases} \qquad E_1 \neq E_2.$$

Assume that the bar has constant cross-sectional area A, is fixed at $x = 0$, and is subjected to an axial end load P at $x = L$. The axial stress is constant throughout the bar:

$$\sigma_{xx} = \frac{P}{A}.$$

From Hooke's law,

$$\sigma_{xx} = E\varepsilon_{xx},$$

the strain is

$$\varepsilon_{xx}(x) = \begin{cases} \dfrac{P}{AE_1}, & 0 < x < a, \\ \dfrac{P}{AE_2}, & a < x < L. \end{cases}$$

Thus,

$$\varepsilon_{xx}(a^-) = \frac{P}{AE_1} \neq \frac{P}{AE_2} = \varepsilon_{xx}(a^+),$$

so the strain has a jump at the material interface.

Within each material,

$$\varepsilon_{xx} = \frac{\mathrm{d}u}{\mathrm{d}x}.$$

Using $u(0) = 0$ and displacement continuity at $x = a$, the displacement field is

$$u(x) = \begin{cases} \dfrac{P}{AE_1}x, & 0 \leq x \leq a, \\ \dfrac{P}{AE_1}a + \dfrac{P}{AE_2}(x-a), & a < x \leq L. \end{cases}$$

The displacement is continuous at the material interface:

$$u(a^-) = u(a^+) = \frac{Pa}{AE_1}.$$

However, its one-sided classical derivatives are

$$\left.\frac{\mathrm{d}u}{\mathrm{d}x}\right|_{x=a^-} = \frac{P}{AE_1}, \qquad \left.\frac{\mathrm{d}u}{\mathrm{d}x}\right|_{x=a^+} = \frac{P}{AE_2}.$$

Since $E_1 \neq E_2$, these two values are different. Therefore, the classical derivative of u does not exist at $x = a$.

Define

$$w(x) = \begin{cases} \dfrac{P}{AE_1}, & 0 < x < a, \\ \dfrac{P}{AE_2}, & a < x < L. \end{cases}$$

We claim that w is the weak derivative of u. Physically,

$$w = \varepsilon_{xx}.$$

To prove the claim, let $\varphi \in C_0^\infty(0, L)$. We must show that

$$\int_0^L w\varphi \,\mathrm{d}x = -\int_0^L u\frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x.$$

Split the right-hand side at the material interface:

$$-\int_0^L u\frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x = -\int_0^a u\frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x - \int_a^L u\frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x. \tag{5}$$

On each material subdomain, u is classically differentiable. Using integration by parts,

$$\begin{aligned} -\int_0^a u\frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x &= -\left[u\varphi\right]_0^a + \int_0^a \frac{\mathrm{d}u}{\mathrm{d}x}\varphi\,\mathrm{d}x \\ &= -u(a^-)\varphi(a) + u(0)\varphi(0) + \int_0^a \frac{P}{AE_1}\varphi\,\mathrm{d}x, \end{aligned} \tag{6}$$

and

$$-\int_a^L u\frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x = -\left[u\varphi\right]_a^L + \int_a^L \frac{\mathrm{d}u}{\mathrm{d}x}\varphi\,\mathrm{d}x$$

$$= -u(L)\varphi(L) + u(a^+)\varphi(a) + \int_a^L \frac{P}{AE_2}\varphi\,\mathrm{d}x. \tag{7}$$

Since $\varphi \in C_0^\infty(0,L)$,

$$\varphi(0) = \varphi(L) = 0.$$

Moreover, displacement continuity at the material interface gives

$$u(a^-) = u(a^+).$$

Therefore, the interface terms cancel:

$$-u(a^-)\varphi(a) + u(a^+)\varphi(a) = 0.$$

Adding (6) and (7) gives

$$-\int_0^L u\frac{\mathrm{d}\varphi}{\mathrm{d}x}\,\mathrm{d}x = \int_0^a \frac{P}{AE_1}\varphi\,\mathrm{d}x + \int_a^L \frac{P}{AE_2}\varphi\,\mathrm{d}x$$

$$= \int_0^L w\varphi\,\mathrm{d}x. \tag{8}$$

Hence, w is the weak derivative of u, and

$$\boxed{\frac{\mathrm{d}u}{\mathrm{d}x} = \varepsilon_{xx} = \begin{cases} \dfrac{P}{AE_1}, & 0 < x < a, \\ \dfrac{P}{AE_2}, & a < x < L, \end{cases} \qquad \text{a.e. in } (0,L).}$$

The value assigned to the weak derivative at the single point $x = a$ is irrelevant.

Thus, the displacement u is continuous but is not classically differentiable at the material interface. Nevertheless, it possesses a weak derivative, and this weak derivative is precisely the piecewise-defined strain ε_{xx}. This example illustrates why weak derivatives arise naturally in mechanics when material properties are discontinuous.